



Badminton Court Management System

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1. Introduction

1.1 Background and Context

In the contemporary landscape of organisational management, the discipline of systems analysis has emerged as a cornerstone of digital transformation initiatives. Systems analysis provides a structured methodological framework for examining existing operational procedures, identifying inefficiencies, and designing optimised information systems that align with organisational objectives. The rapid growth of web and mobile technologies has significantly influenced how organizations operate, encouraging a shift from manual processes toward automated and data-driven solutions.

The Pemsiri Indrajith Memorial Badminton Indoor Stadium is a dedicated badminton-only facility that manages a limited number of indoor courts and serves a growing base of players. As a specialised sports venue focused entirely on badminton, the stadium must efficiently coordinate court reservations, membership handling, and communication with users. However, the management of a small-scale facility with only two courts presents unique challenges, particularly in balancing high demand with limited availability. The reliance on manual or semi-digital processes such as phone bookings, walk-in reservations, and spreadsheet-based record keeping makes it difficult to manage these constraints effectively.

The finite availability of just two courts, combined with fluctuating user demand and the need for timely communication, creates an environment where manual management approaches are increasingly inadequate. Even minor inefficiencies, such as delayed updates or missed communication, can lead to booking conflicts, underutilization of courts, and reduced customer satisfaction. This highlights the importance of applying systems analysis principles to identify inefficiencies and propose a structured, technology-driven solution.

This report presents a comprehensive systems analysis conducted for the Pemsiri Indrajith Memorial Badminton Indoor Stadium. The analysis aims to identify limitations in the current semi-manual system and define the requirements for a modern Badminton Club Management System. The proposed system focuses on improving operational efficiency through features

such as real-time booking management and automated notification mechanisms to ensure optimal utilization of the limited court resources.

1.2 Project Initiation

The project was initiated in response to operational challenges observed at the Pemsiri Indrajith Memorial Badminton Indoor Stadium, which operates as a small-scale indoor badminton facility with only two courts. The stadium currently manages its daily activities using a combination of telephone bookings, walk-in reservations, and basic spreadsheet tools. While this approach may have been sufficient during earlier stages of operation, the increasing number of users has exposed significant inefficiencies in the system.

Initial investigations revealed frequent issues such as booking conflicts, delays in confirming reservations, and difficulties in managing cancellations. Due to the absence of a real-time notification mechanism, cancelled time slots often remain unutilized, leading to lost opportunities for both players and management. Given the limited number of courts, even a small number of unfilled slots has a noticeable impact on overall court utilization and revenue generation.

Additionally, the manual coordination process increases the likelihood of human errors such as double bookings and miscommunication between staff and users. These challenges not only affect operational efficiency but also reduce user satisfaction and trust in the system. These findings served as the primary motivation for proposing the development of an integrated Badminton Club Management System that can automate booking processes, reduce errors, and improve communication through real-time updates and notifications.

The project is being developed as part of an academic initiative, applying systems analysis and software engineering principles to address real-world operational challenges. It focuses on designing a practical and scalable solution tailored specifically to the needs of a badminton-only facility with limited court resources.

1.3 Strategic Importance

The strategic importance of the proposed Badminton Club Management System lies in its ability to transform the operational efficiency of the Pemsiri Indrajith Memorial Badminton

Indoor Stadium while addressing the constraints of managing only two courts. By introducing a centralized and automated system, the stadium can significantly reduce administrative workload and improve the accuracy of booking and membership management processes.

One of the key strategic advantages of the system is its ability to maximize court utilization. With only two courts available, efficient scheduling and quick reallocation of cancelled slots are critical. The proposed system incorporates real-time notification features that inform users about newly available slots, thereby reducing idle court time and minimizing revenue loss. This capability provides a competitive advantage by ensuring that limited resources are used as effectively as possible.

Furthermore, the system supports data-driven decision-making through the collection and analysis of operational data. Insights into peak usage times, cancellation patterns, and user behavior can help management optimize scheduling, improve service quality, and plan future enhancements. The centralized database structure ensures data consistency, accessibility, and reliability, which are essential for effective management.

From a long-term perspective, the system establishes a scalable digital foundation that can accommodate future growth in membership without requiring proportional increases in administrative effort. It also enhances the overall user experience by providing a more convenient, transparent, and reliable booking process. In an environment where efficiency and user satisfaction are critical, the implementation of this system represents a significant step toward modernizing the stadium's operations and ensuring its sustainability.

2. Fact-Finding

2.1 Justification of Fact-Finding Techniques

The selection of fact-finding techniques for the Pemsiri Indrajith Memorial Badminton Indoor Stadium was based on the small-scale, semi-manual nature of its operations with only two courts. Structured interviews were used to gather detailed insights from key individuals like the club owner and coaching coordinator, as most operational knowledge is not formally documented. Questionnaires were chosen to collect feedback from members efficiently and

anonymously, helping to understand user behavior and expectations. Direct observation was applied to capture real-time system operations and identify practical issues that may not be revealed through interviews alone. Other techniques such as document analysis and focus groups were not suitable due to the lack of formal records and the difficulty of organizing participants, making the selected methods the most appropriate for accurate and reliable data collection..

2.1.1 Structured Interviews

Structured interviews were selected as the primary fact-finding method due to the semi-manual nature of the current system and the need to gather detailed insights from key personnel involved in daily operations. In a small facility such as this stadium, operational knowledge is largely experience-based and not formally documented. Staff members, including the manager and front-desk personnel, handle bookings, cancellations, and scheduling based on routine practices and personal judgment rather than standardized procedures.

The structured format ensured that consistent information was collected from all interviewees while allowing the analysis to focus on critical operational areas such as booking management, cancellation handling, and communication methods. Given that staff members often have limited time due to continuous customer interaction, structured interviews allowed efficient data collection within a limited timeframe. This method was also suitable because staff may not be familiar with technical system concepts, and a guided set of questions helped extract relevant information without requiring technical knowledge.

Interviews were particularly important for understanding the impact of cancellations and booking conflicts. Since the current system does not maintain detailed digital records, it was necessary to rely on staff estimates and experiences to assess how often issues such as double bookings and unused time slots occur, especially considering the high demand for only two available courts.

2.1.2 Questionnaires

Questionnaires were used to collect information from the users of the stadium, including regular players and registered members. This method was appropriate because it allowed data

collection from a larger group of users within a short period, which would not have been practical through interviews alone. The users of the stadium vary in age, playing frequency, and familiarity with digital systems, making it necessary to use a standardized approach to gather comparable data.

The questionnaire was designed to capture user experiences related to booking methods, cancellation issues, communication effectiveness, and overall satisfaction with the current system. Since some users may hesitate to express negative feedback directly to staff, the anonymous nature of questionnaires encouraged more honest responses. This helped in identifying real user concerns, such as difficulties in securing court bookings, lack of timely updates about availability, and inconvenience caused by manual processes.

Additionally, the questionnaire included simple and clear questions using non-technical language to ensure that all users could respond easily. The collected data provided valuable insights into user expectations and highlighted the need for features such as real-time booking visibility and automated notifications.

2.1.2.1 Questionnaire Structure and Design Rationale

The questionnaire was carefully designed to collect both quantitative and qualitative data relevant to the operational characteristics of the badminton stadium. The structure followed a logical progression, beginning with respondent profiling questions and gradually moving toward operational issues and system improvement expectations.

The questionnaire consisted of the following structured sections:

1. Demographic and Usage Information
2. Booking Behaviour
3. Cancellation Experience
4. System Satisfaction
5. Technology Readiness and Feature Expectations

This structured sequencing ensured that respondents were gradually guided from neutral background questions toward more evaluative and opinion-based responses, thereby reducing response bias and improving reliability.

2.1.3 Direct Observation

Direct observation was employed to gain an accurate understanding of how the current system operates in practice. This technique allowed the analysis to capture real-time activities and identify discrepancies between how processes are intended to work and how they are actually carried out.

Observation was particularly useful in examining the booking and cancellation processes. By observing staff during daily operations, it was possible to document how bookings are received, recorded, and managed using manual methods. The observation also revealed how cancellations are handled, including whether cancelled slots are effectively communicated to other users or left unfilled. Given that the stadium has only two courts, even a single unutilized time slot represents a significant loss in potential usage and revenue.

Observations conducted during peak hours, such as evenings and weekends, provided insights into how the system performs under high demand. It highlighted issues such as delays in processing bookings, increased chances of double booking, and difficulties in maintaining accurate records when multiple requests occur simultaneously.

The use of these three fact-finding techniques ensures a comprehensive and reliable analysis. By combining insights from staff (interviews), users (questionnaires), and real-world operations (observation), the findings can be cross-verified, increasing the validity of the results and providing a strong foundation for defining the requirements of the proposed system.

2.2 Completeness and Quality of Fact-Finding

2.2.1 Interviews Conducted

A total of two structured interviews were conducted over a one-week period at the Pemsiri Indrajith Memorial Badminton Indoor Stadium. Due to the small-scale nature of the facility, which operates with only two courts and a limited number of staff, key operational responsibilities are concentrated among a few individuals. Therefore, interviews were focused on personnel who possess the most comprehensive understanding of the system's daily

operations and decision-making processes. The interview subjects and their roles are presented below:

ID	Role	Responsibilities	Duration
INT-01	Club Manager	Overall operations, financial oversight	45 min
INT-02	Coaching Coordinator	Coaching schedule, court allocation	40 min
INT-03	Reception	Booking management, member queries	35 min
INT-04	Staff	Bookings,payments	30 min

Each interview followed a structured format with a predefined set of questions organized into key areas such as booking procedures, cancellation handling, communication methods, and operational challenges. This approach ensured consistency in data collection while allowing the discussion to remain focused on the most critical aspects of the current system.

Sample interview questions included:

- Can you describe the complete process for handling a court booking from the moment a player contacts the stadium to when the booking is confirmed?
- How are cancellations currently recorded, and what steps are taken to reassign the cancelled time slot?
- How do you inform other players about newly available court slots, if at all?
- How often do booking conflicts or double bookings occur, and how are they resolved?
- How are coaching sessions scheduled alongside regular bookings, given the limitation of only two courts?
- What are the main challenges you face in managing court availability and player communication?
- What improvements would you expect from a computerized management system?

Each session lasted between 30 to 45 minutes and was documented through note-taking, with additional clarification obtained where necessary. The information gathered from these interviews provided valuable insights into the operational limitations of the current system, particularly in managing bookings and cancellations within a highly constrained two-court environment.

2.2.2 Questionnaire Design and Distribution

A structured questionnaire consisting of 22 questions was developed and distributed to players of the Pemsiri Indrajith Memorial Badminton Indoor Stadium using a Google Form. The questionnaire was designed to gather player perspectives on the current system and identify expectations for the proposed management system. It was divided into five sections: demographic information (4 questions), booking behaviour (5 questions), cancellation experience (4 questions), notification preferences (5 questions), and system feature expectations (4 questions). A combination of multiple-choice questions, Likert scale ratings, and simple ranking questions was used to collect both quantitative and qualitative data.

Due to the small and focused user base of the stadium, the questionnaire was shared with approximately 20 active players who regularly use the facility. The Google Form link was distributed digitally, making it easy for players to respond at their convenience using mobile devices. This method ensured quick data collection and improved response accuracy.

Out of the 20 distributed questionnaires, a majority of responses were received, providing sufficient data to identify common patterns and user concerns within this small-scale environment. Although the sample size is limited, it is appropriate given that the stadium operates only two courts and serves a consistent group of regular players. The responses provided valuable insights into booking habits, issues faced during peak hours, dissatisfaction with current communication methods, and a strong preference for features such as real-time availability updates and automated notifications.

The findings from the questionnaire complemented the data gathered from interviews and observation, contributing to a comprehensive understanding of the current system and supporting the identification of key requirements for the proposed solution.

2.2.3 Observation Scope and Methodology

Direct observation was conducted to understand the real-world operation of the system at the Pemsiri Indrajith Memorial Badminton Indoor Stadium. Information was primarily gathered through physical visits to the stadium, where discussions were held with the club owner and coaching coordinator to understand booking processes, cancellation handling, and daily

operations. In addition to staff interactions, player-related information was collected using a Google Form questionnaire to capture user experiences and preferences.

The observation focused on key operational areas including booking procedures, cancellation handling, court usage during peak hours, and staff–player interactions. Rather than following a rigid time-based observation schedule, the analysis was based on multiple visits and real-time discussions, allowing flexibility in understanding different scenarios and operational challenges. Notes were taken during these interactions to document process flows, identify inefficiencies, and observe how tasks are handled in practice.

This combined approach of physical observation and digital data collection ensured a comprehensive understanding of the system from both staff and user perspectives. It also helped validate findings across different sources, improving the reliability of the analysis.

Appropriateness of Observations:

To ensure the quality of the fact-finding process, we have to consider some appropriateness of the results that we obtained from our questionnaire. The following considerations reflect the Appropriateness of the observations.

Each question was specifically designed to address known operational weaknesses of the current system rather than relying on generic satisfaction questions.

For example:

- Questions regarding booking method (phone, walk-in, staff) directly investigated the dependency on manual processes.
- Questions about booking difficulties measured usability limitations.
- Cancellation-related questions targeted the core problem of revenue loss due to unutilized cancelled slots.
- Real-time notification questions directly tested feasibility and acceptance of the proposed innovative feature.

This ensured that all responses were directly aligned with the system development objectives.

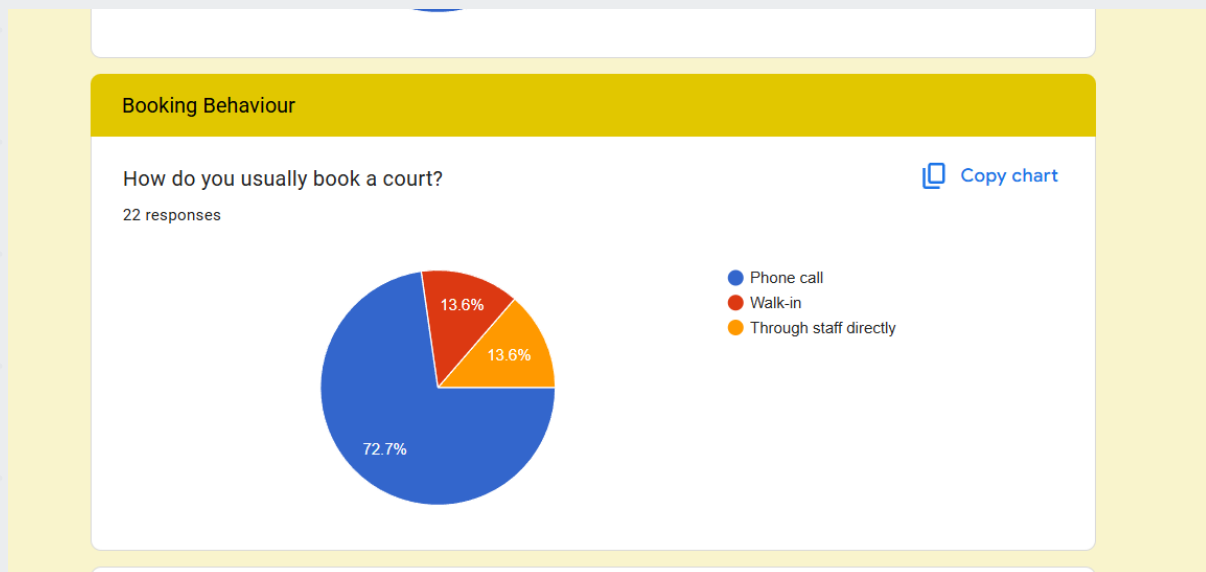


Figure 1 responses

Booking Method

- **72.73% book via phone call**
- 13.64% through staff directly
- 13.64% walk-in

This confirms that the current system is heavily dependent on manual communication channels.

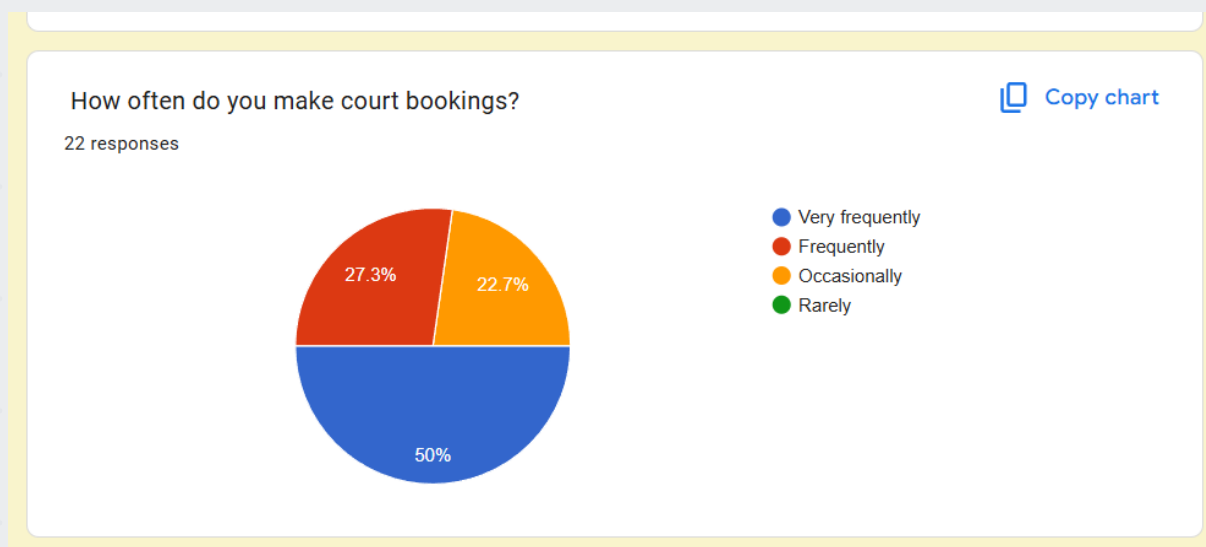


Figure 2 Responses

Booking Frequency

- **50% book very frequently**
- 27.27% book frequently
- 22.73% occasionally

Frequent booking activity strengthens the need for automation.

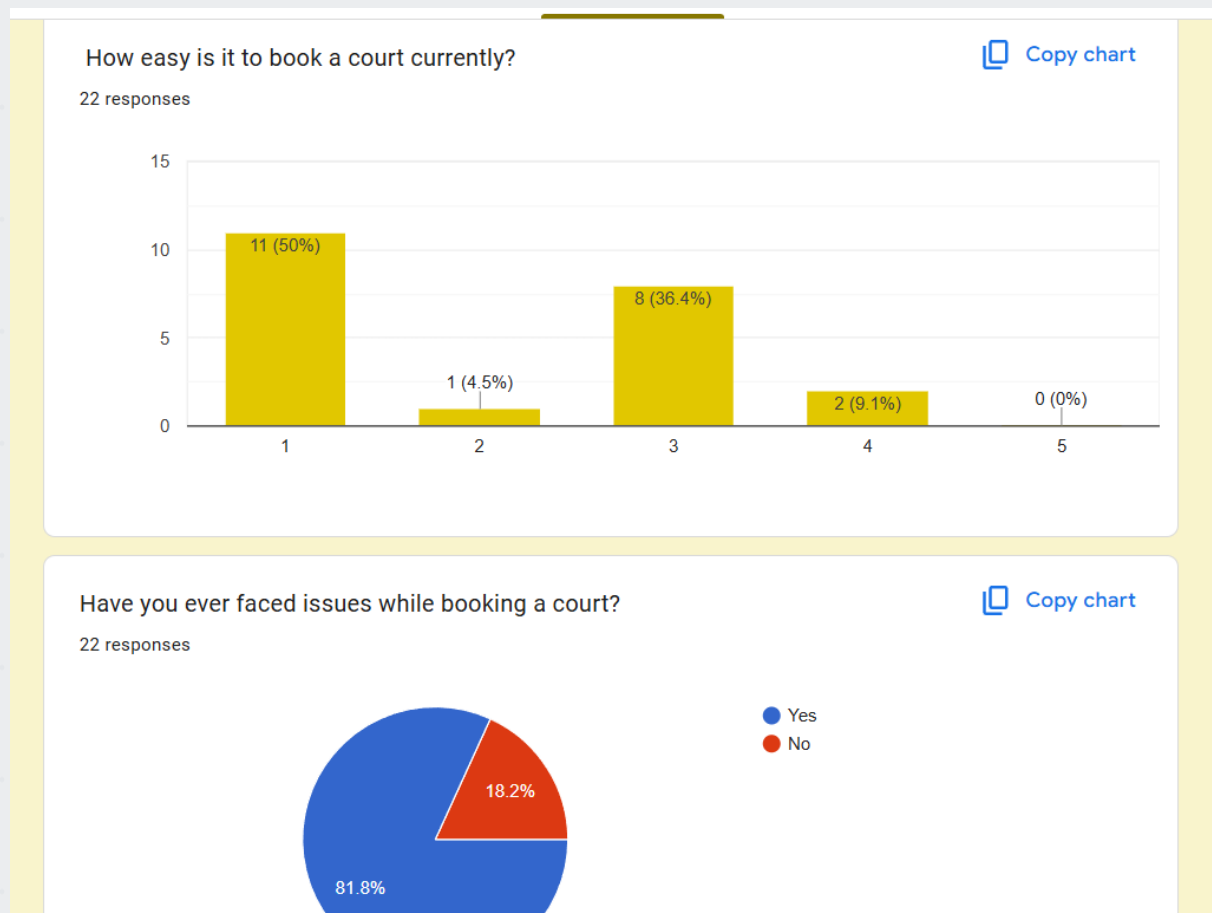


Figure 3 Responses

Ease of Booking (Scale 1–5)

- **50% rated 1 (Very difficult)**
- 36.36% rated 3
- Only 9.09% rated 4

This shows strong dissatisfaction with the booking process.

Booking Issues

- **81.82% have faced issues while booking**
- Only 18.18% have not

This is a critical indicator of operational inefficiency.

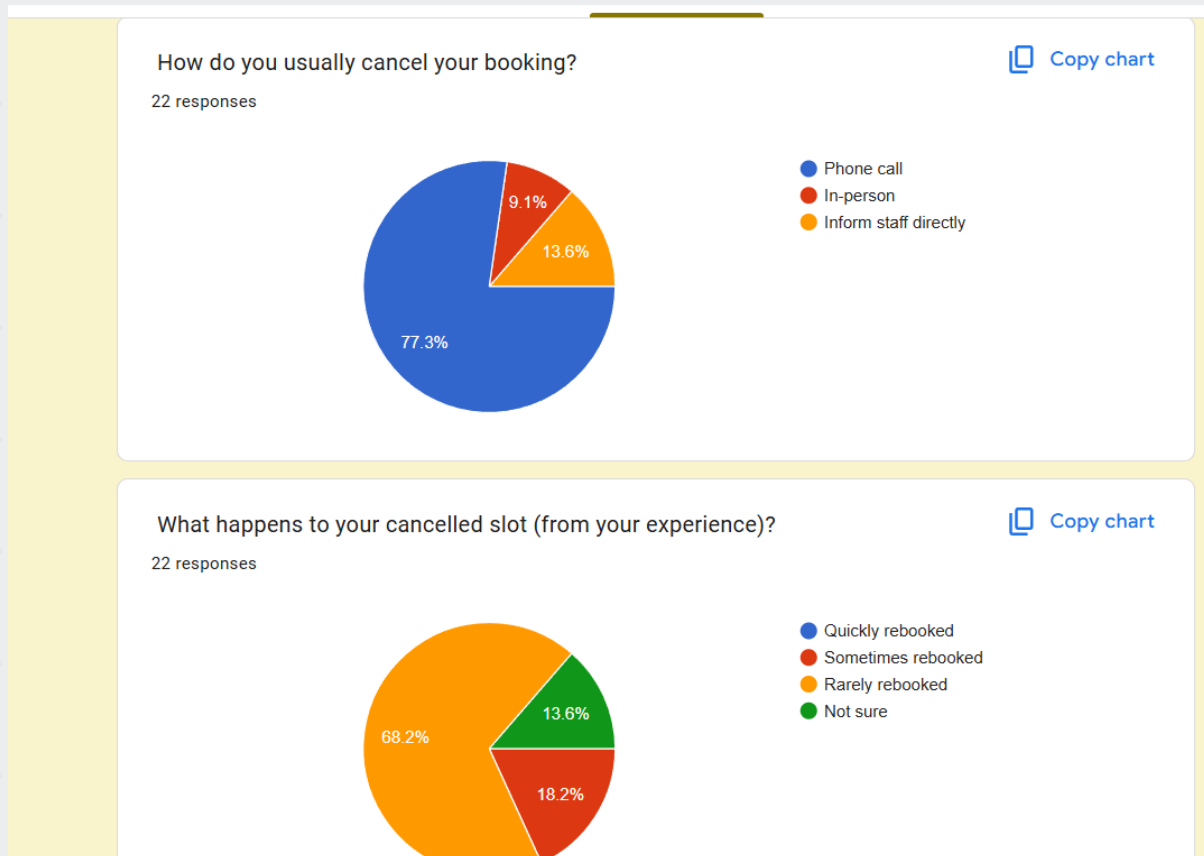


Figure 4 Responses

Cancellation Method

- **77.27% cancel via phone**
- 13.64% inform staff
- 9.09% cancel in person

This confirms absence of digital cancellation features.

Rebooking of Cancelled Slots

- **68.18% say cancelled slots are rarely rebooked**
- 18.18% say sometimes rebooked
- 13.64% unsure

This strongly supports the revenue-loss problem identified in interviews.

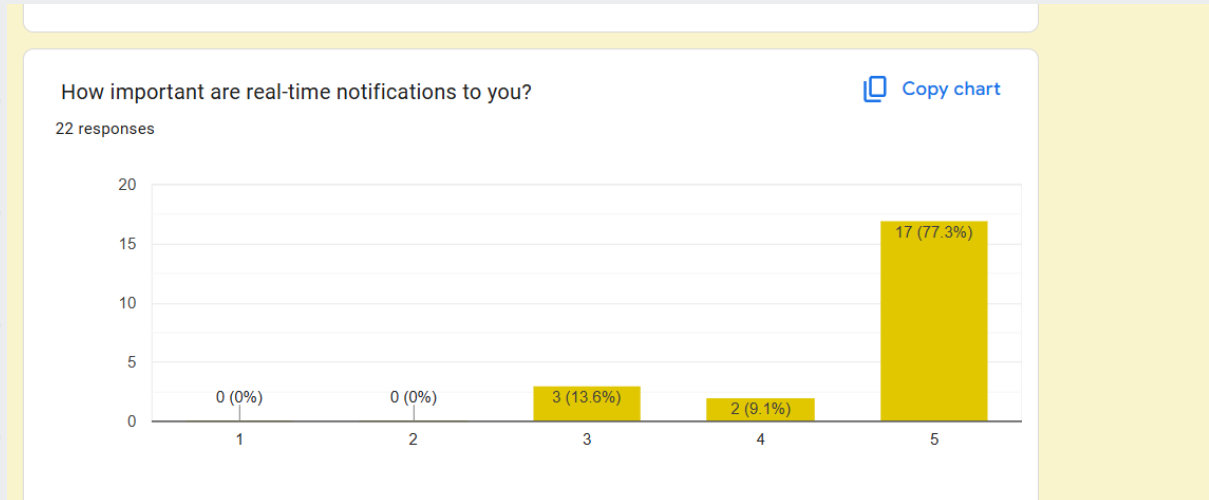


Figure 5 Responses

Importance of Real-Time Notifications

- **77.27% rated importance as 5 (Very Important)**
- 13.64% rated 3
- 9.09% rated 4

This provides strong quantitative justification for the real-time cancellation alert feature.

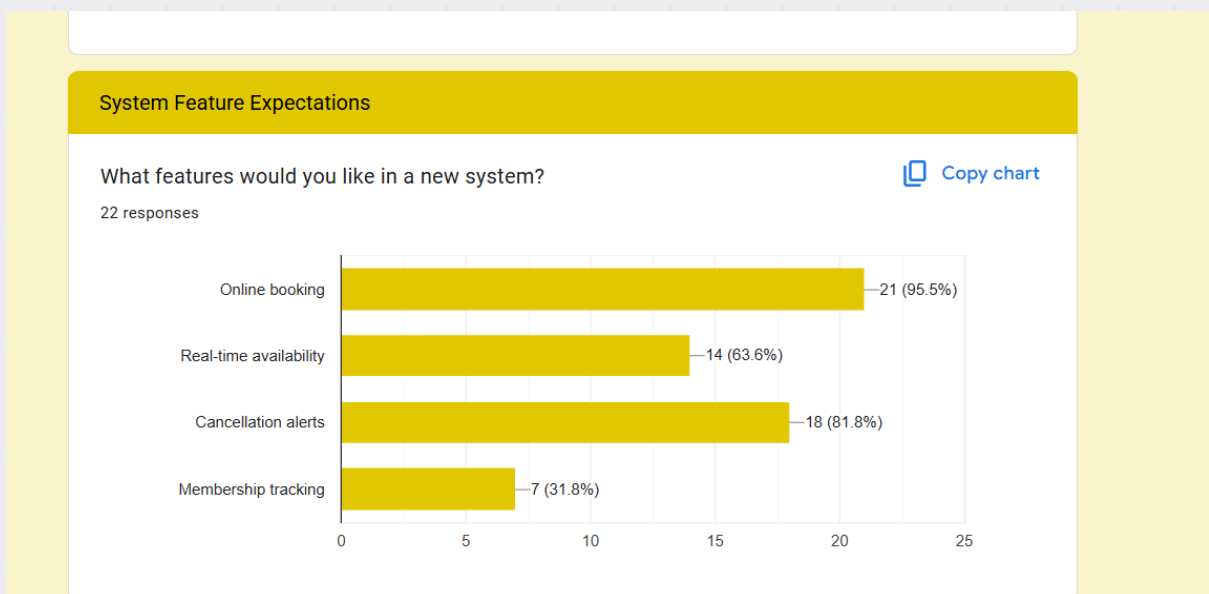


Figure 6 Responses

Desired Features in New System

Most common requested features include:

- Online booking
- Real-time availability
- Cancellation alerts
- Membership tracking

Notably:

- 27.27% selected all four major features.
- 27.27% selected booking + availability + alerts.
- 27.27% selected booking + alerts.

This shows strong demand specifically for:

1. Online booking
2. Real-time visibility
3. Cancellation alerts

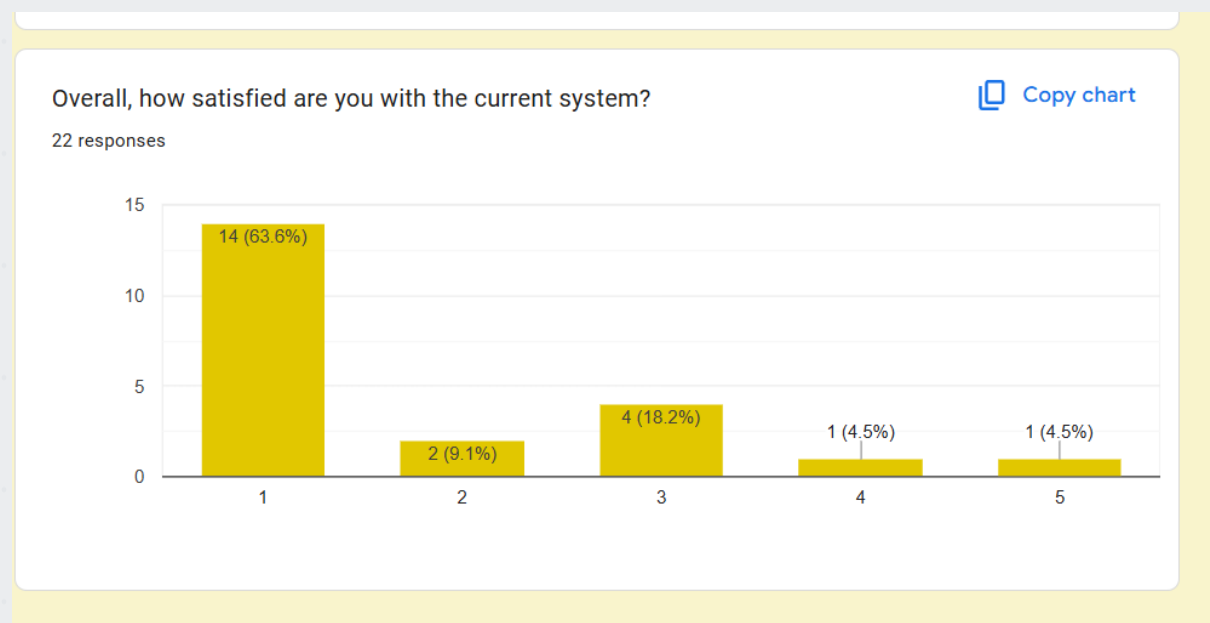


Figure 7 Responses

Overall System Satisfaction

- **63.64% rated overall satisfaction as 1 (Very dissatisfied)**
- 18.18% rated 3
- Only 9% rated 4 or 5

This indicates significant dissatisfaction with the current system.

Conclusive Observations

From the consolidated findings:

1. The current manual booking system is inefficient and highly error-prone.
2. Cancellation handling lacks real-time reallocation mechanisms.
3. A significant proportion of cancelled slots remain unused.
4. Users are dissatisfied with booking and cancellation processes.
5. There is strong demand for:
 - Online booking
 - Real-time cancellation alerts
 - Automated reminders
 - Mobile application access
6. The user base is digitally ready and technologically capable of adopting the new system.

2.3 Analysis of Results

2.3.1 Booking and Cancellation Patterns

The findings gathered from player questionnaires and interviews with the owner and staff of the Pemsiri Indrajith Memorial Badminton Indoor Stadium highlight several inefficiencies in the current booking and cancellation process. Since the stadium operates with only two courts, effective management of bookings is critical. However, the existing system, which relies on phone calls, walk-ins, and direct communication with staff, is not efficient enough to handle user demand.

According to questionnaire responses, the majority of users are regular players who book courts frequently, with many playing at least once per week. A significant number of users prefer peak hours, particularly weekday evenings and extended hours during weekends. During these periods, booking requests increase considerably, making it difficult for staff to manage reservations accurately using manual methods. As confirmed through staff interviews, this often leads to issues such as booking conflicts, missed entries, and occasional double bookings.

The analysis also shows that cancellations occur regularly. Based on staff feedback, there are approximately 5–8 cancellations per week, most of which occur between Friday and Sunday.

The average notice period for cancellations is around 4–6 hours, as indicated by both questionnaire responses and interviews. Despite this, the current system lacks a structured approach to handle cancellations effectively.

Metric	Value	Source
Average cancellations per week	5-8	Staff interviews
Cancellations unbooked	60%	Observation + Staff estimate
Peak cancellation days	Friday-Sunday	Observation data
Average notice period	4-6 hours	Questionnaire + Interviews
Members wanting cancellation alerts	82%	Questionnaire (n=20)

When a booking is cancelled, staff manually update records, but there is no proactive mechanism to inform other players about newly available time slots. As a result, a large proportion of cancelled bookings remains unused. In a facility with only two courts, each unutilized time slot represents a direct loss in potential usage and revenue. Currently, rebooking depends on chance factors such as walk-in players or incoming phone calls rather than a systematic process.

Furthermore, both interview and questionnaire data clearly indicate that players are interested in receiving notifications about available slots. The absence of such a feature is a significant limitation of the current system. Overall, these findings highlight the need for a real-time booking and notification mechanism to improve court utilization, reduce operational inefficiencies, and enhance overall user satisfaction.

2.3.2 Double-Booking Incidents

Double-booking was identified as a major operational issue across all fact-finding activities conducted at the Pemsiri Indrajith Memorial Badminton Indoor Stadium. Both interview responses and direct observation confirmed that booking conflicts occur regularly, particularly during peak hours within the stadium's operating periods (weekdays 14:00–23:00 and weekends 06:00–23:00), when demand for the two available courts is highest. The primary cause of this issue is the reliance on manual record-keeping methods, where bookings are recorded in spreadsheets or logs with a time delay between receiving a request and updating the system.

In situations where multiple booking requests are handled simultaneously—such as a phone reservation occurring while a walk-in booking is being processed—the same court and time slot may be unintentionally assigned to more than one player. This lack of real-time synchronization creates a high risk of overlapping bookings, especially during busy periods when staff must respond quickly.

Factor	Frequency	Impact
Double-bookings per week	2-3 incidents	High
Peak period incidents	4-5 per weekend	High
Average resolution time	10-15 minutes	Medium
Compensation cost per incident	Discount/free booking	Financial
Member complaints per month	4-6 formal complaints	Reputational

The consequences of double-booking extend beyond immediate inconvenience to players. When such conflicts occur, staff must spend additional time resolving the issue by negotiating alternative time slots or offering compensation. This leads to financial loss and disrupts overall operations, particularly during high-demand periods.

Furthermore, repeated booking conflicts reduce user trust and satisfaction. Players who experience such issues may lose confidence in the reliability of the current system, which can negatively impact long-term engagement with the facility. Given that the stadium operates with only two courts, even a small number of errors can significantly affect overall performance. These findings clearly highlight the need for a real-time, centralized booking system to prevent overlapping reservations and ensure accurate scheduling.

2.3.3 Member Communication and Notification Preferences

Analysis of questionnaire responses collected from members of the Pemsiri Indrajith Memorial Badminton Indoor Stadium indicates a strong preference for mobile-based communication methods when receiving updates about court availability and cancellations. Given the small but active user base and the high demand for only two courts, timely communication is critical for improving court utilization and reducing idle time.

When members were asked how they would prefer to be notified about cancelled or newly available court slots, the responses were as follows:

Notification Channel	Member Preference (n=20)
SMS (Text Message)	78%
WhatsApp Message	71%
Mobile App Push Notification	65%
Email Notification	42%
Phone Call	8%

The results clearly show that SMS and WhatsApp are the most preferred communication channels among members. This reflects the behavior of users who rely on mobile messaging for quick and convenient updates, especially for time-sensitive activities such as court bookings. Email is less effective for urgent communication, while phone calls are the least preferred due to inconvenience and time consumption.

These findings strongly support the need for a multi-channel notification system in the proposed solution. By integrating SMS and WhatsApp notifications, the system can instantly inform members about newly available slots, increasing the likelihood of rebooking cancelled sessions. This is particularly important in a facility with only two courts, where efficient communication directly impacts utilization and revenue.

Overall, the analysis highlights that effective and real-time communication is a key requirement for the new system, and the adoption of mobile-based notification methods will significantly enhance both user satisfaction and operational efficiency.

2.3.4 Membership Management Deficiencies

The analysis of membership management practices at the Pemsiri Indrajith Memorial Badminton Indoor Stadium reveals several inefficiencies resulting from the current semi-manual approach. Due to the small scale of the facility, membership records are maintained using basic spreadsheets and are manually updated when new members register or renew. While this method is manageable at a minimal level, it lacks automation, accuracy, and consistency, leading to administrative challenges.

A key issue identified is the absence of a systematic mechanism to track membership expiry dates. Currently, staff are responsible for manually monitoring renewals and informing members. However, this process is inconsistent and often overlooked due to daily operational workload. As a result, memberships frequently lapse without timely renewal, negatively affecting both member engagement and potential revenue.

Interview findings confirm that membership tracking is not strictly enforced, while questionnaire responses highlight user dissatisfaction. A considerable number of members are unaware of their membership status or expiry dates, and many expressed the need for a more reliable and transparent system. Additionally, there is strong interest among members in receiving automated reminders prior to membership expiry, indicating a clear gap in the current communication process.

The lack of an automated membership management system leads to several consequences, including missed renewal opportunities, reduced member retention, and increased administrative effort. These issues demonstrate the need for a centralized solution capable of maintaining accurate member records, tracking expiry dates, and automatically notifying members about upcoming renewals.

Overall, the findings strongly support the inclusion of a dedicated membership management module in the proposed system, incorporating features such as automated reminders, real-time status tracking, and improved data organization to enhance both operational efficiency and user experience.

2.3.6 Key Conclusive Observations

Based on the comprehensive analysis of all fact-finding activities conducted at the Pemsiri Indrajith Memorial Badminton Indoor Stadium, the following key observations are identified:

- The current semi-manual booking system is inefficient and difficult to manage, especially with only two courts available. Booking conflicts such as double-bookings occur regularly (approximately 1–2 times per week), and a significant amount of staff time is spent handling manual records and resolving these issues.

- The lack of a real-time cancellation notification mechanism results in many cancelled court slots remaining unused. A considerable portion of cancellations are not rebooked, leading to reduced court utilization and missed opportunities for both players and management.
- Member preferences clearly indicate a strong demand for mobile-based communication, particularly through SMS and WhatsApp. This supports the need for a multi-channel notification system to provide timely updates about bookings and cancellations.
- The current membership management process is manual and lacks proper tracking and reminder features. This leads to missed renewals, reduced member engagement, and unnecessary administrative workload that could be minimized through automation.
- There is no centralized reporting or data management system in place, making it difficult for management to analyze usage patterns, monitor performance, or make informed decisions based on accurate data.

These observations highlight the key weaknesses of the current system and establish a clear foundation for defining the requirements of the proposed Badminton Club Management System. They directly support the need for a centralized, automated solution that improves efficiency, reduces errors, and enhances overall user experience.

3. Description of Current System

3.1 Narrative Description

3.1.1 System Overview

The current management system at the Pemsiri Indrajith Memorial Badminton Indoor Stadium operates as a semi-manual information system that relies on telephone-based booking coordination, walk-in reservation handling, spreadsheet-based record keeping, and informal communication practices. As a badminton-only facility with only two courts, the system has evolved gradually without formal design or documentation. This has resulted in a set of loosely connected processes that depend heavily on staff experience and manual coordination rather than an integrated digital platform. Due to the lack of real-time synchronization and centralized control, the system is prone to inefficiencies, especially during peak usage periods.

3.1.2 System Inputs

The system receives inputs from multiple sources through different operational channels. The primary inputs include:

- **Booking Requests:** Players submit court booking requests either through telephone calls or by visiting the stadium. Each request includes details such as date, time, and number of players.
- **Cancellation Notifications:** Players cancel bookings by informing staff via phone or in person. Most cancellations occur on the same day or a few hours before the scheduled time.
- **Membership Applications:** New members provide their personal details through manual forms, while existing members request renewals through direct interaction with staff.

3.1.3 System Processes

The core processes of the system are carried out manually by staff using a combination of verbal communication and simple record-keeping tools. The main processes include:

1. **Booking Processing:** Staff check availability using a spreadsheet or manual record. If a slot is free, booking details are entered manually and confirmed verbally or through a receipt.
2. **Cancellation Processing:** When a cancellation occurs, staff update the record manually. However, the available slot is not actively communicated to other players.
3. **Double-Booking Resolution:** If two players are assigned the same time slot, staff must resolve the issue manually by negotiating alternative arrangements, sometimes offering compensation.
4. **Membership Management:** Member details are recorded and updated manually. Expiry dates are tracked informally, and no automated reminders are provided.

3.1.4 System Outputs

The current system produces limited outputs, mainly focused on immediate operational needs rather than long-term analysis or decision-making:

- **Booking Confirmations:** Provided verbally for phone bookings or through simple handwritten receipts.
- **Schedules:** Court bookings and tournament schedules are displayed informally, often using written records or boards.
- **Membership Records:** Member information is stored in spreadsheets and accessed when needed, without automated reporting features.

3.2 Illustration Supporting Narrative

3.3 Data Flow Diagrams

3.3.1 Context Diagram (Level 0 DFD)

The Context Diagram for the current system at the Pemsiri Indrajith Memorial Badminton Indoor Stadium represents the system as a single central process interacting with key external entities. The primary external entities identified are Players (Members), the Club Owner, Staff, and Coaching Staff.

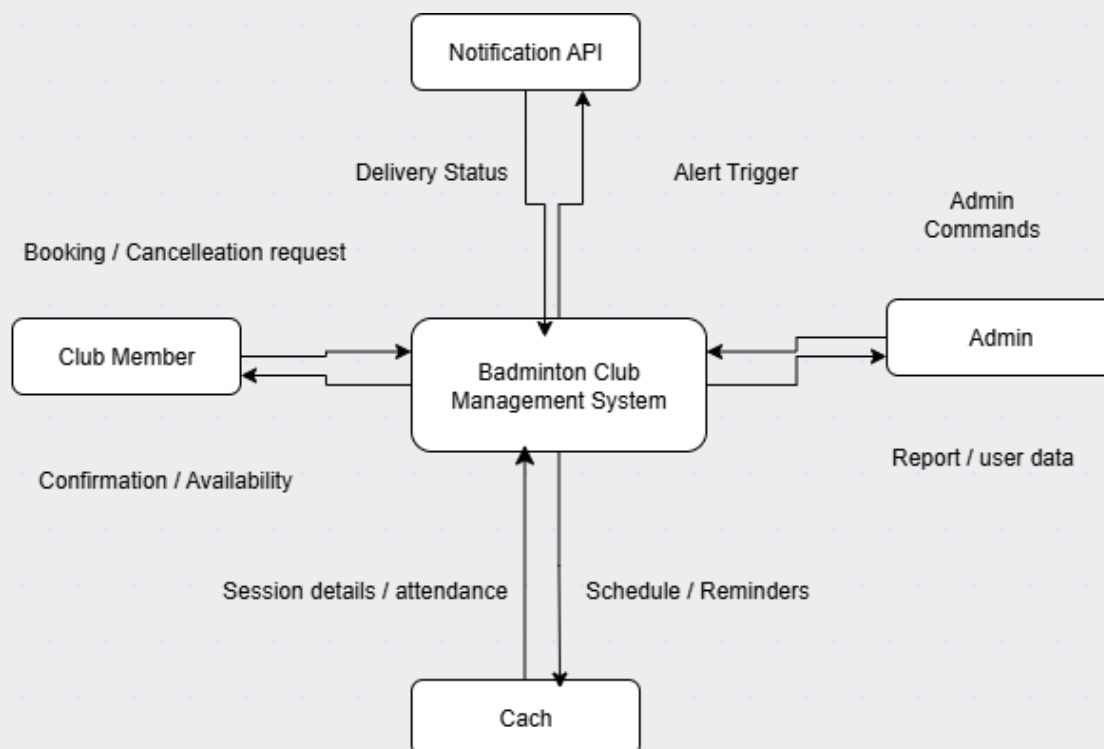
The system receives data flows such as booking requests, cancellation notifications, and membership information from players. Staff and the club owner are responsible for handling

these requests and updating records manually. Coaching staff may also provide scheduling-related information when arranging training sessions.

In return, the system produces outputs including booking confirmations, court availability information, and basic schedules, which are communicated to players mainly through verbal interaction or direct contact. Staff play a key role in managing these outputs by coordinating between players and the owner.

The context diagram illustrates that the current system operates as a single unified process managing all activities, even though in reality it is performed through manual methods such as phone calls, face-to-face communication, and spreadsheet-based records. This abstraction highlights the heavy dependence on human involvement, particularly staff and the owner, rather than an automated system.

Overall, the diagram reflects a simple system structure but also exposes its key limitations, including lack of automation, absence of real-time updates, and inefficient communication between the system and its users.



8 Data flow Context Diagram

3.3.2 Level 1 DFD

The Level 1 Data Flow Diagram decomposes the central process of the system at the Pemsiri Indrajith Memorial Badminton Indoor Stadium into key sub-processes that reflect its primary operational functions. Given the simplicity of the system and the focus on badminton court management, the following core processes are identified:

- **Process 1.0 – Manage Bookings:**

Receives booking requests and cancellation notifications from players through the owner or staff. Booking details are recorded and updated manually in a spreadsheet or log. This process produces booking confirmations and provides court availability information.

- **Process 2.0 – Manage Memberships:**

Receives membership applications and renewal requests from players. Member details are manually recorded and maintained in a spreadsheet. This process manages membership information but does not provide automated tracking or reminders.

- **Process 3.0 – Handle Cancellations and Reallocation:**

Processes booking cancellations received from players. Updates records accordingly and attempts to reassign available slots. However, due to the lack of a structured communication system, reallocation is inconsistent and often depends on manual effort by the owner or staff.

- **Process 4.0 – Manage Court Scheduling:**

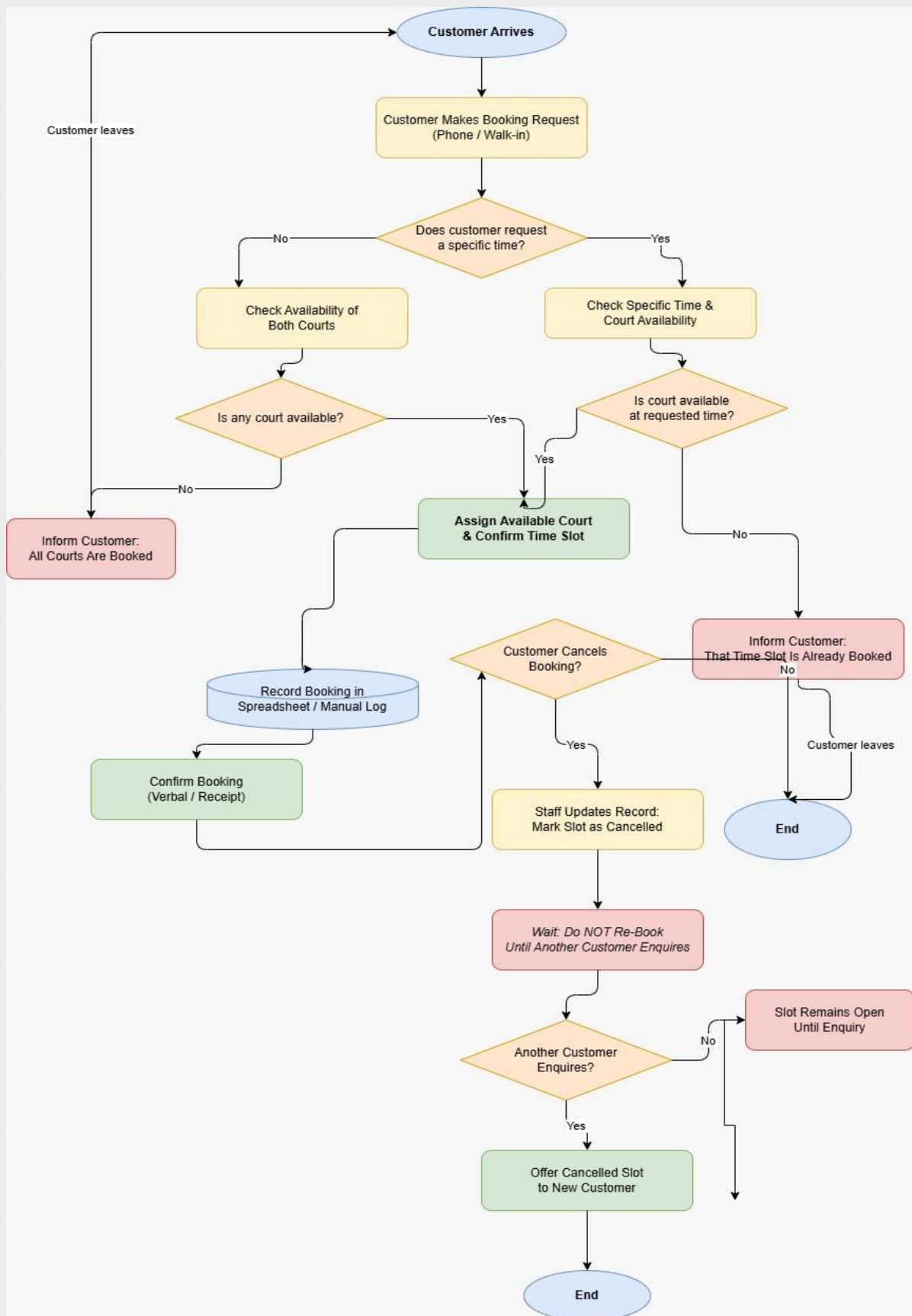
Coordinates the allocation of courts, including permanent bookings and temporary reservations. Ensures that court usage is recorded, although conflicts may occur due to lack of real-time updates.

The data stores identified in the Level 1 DFD include:

- **Booking Records:** Maintained in spreadsheets or manual logs, containing details of all reservations.

- **Member Records:** Stored in spreadsheets with basic member information and status.
- **Court Schedule Records:** Maintained to track permanent bookings and daily court allocations.

Each of these data stores corresponds to manual or semi-digital storage methods currently used within the system. Detailed Level 2 Data Flow Diagrams for each subprocess can be included in the appendix for further explanation.



9 Process Flow Current System

3.4 Problems and Limitations

3.4.1 Operational Inefficiencies

The current system at the Pemsiri Indrajith Memorial Badminton Indoor Stadium suffers from several operational inefficiencies that negatively affect service quality and overall management effectiveness. The booking process is entirely manual, requiring players to contact the owner or staff via phone or in person. On average, each booking takes approximately 3–5 minutes under normal conditions, but this can increase during peak periods when multiple players request bookings at the same time. Since the system depends heavily on one person (the owner) or limited staff, handling multiple requests simultaneously becomes difficult and time-consuming.

The absence of real-time availability information creates a major limitation. Players cannot independently check available time slots and must rely on the owner or staff to provide this information. This often leads to delays, miscommunication, and missed booking opportunities. Additionally, since permanent bookings occupy a large portion of court time, identifying available slots for non-regular players becomes even more complex without a structured system.

Cancellation handling is the most critical inefficiency in the current system. When a booking—especially a permanent booking—is cancelled, the owner or staff must manually update records and attempt to inform other players. However, due to the lack of a proper communication mechanism and limited access to player contact information, many cancelled slots are not reassigned. As a result, a significant percentage of available court time remains unused, even when there is demand from other players.

Furthermore, the system lacks coordination between booking updates and player communication. Available slots are often not matched efficiently with interested players, leading to underutilization of the facility. Given that the stadium operates with only two courts, these inefficiencies have a direct impact on both court usage and user satisfaction.

3.4.2 Financial Impact

- The financial impact of the current system deficiencies at the Pemsiri Indrajith Memorial Badminton Indoor Stadium can be observed through interview findings and

analysis of daily operations. Although the facility operates on a small scale with only two courts, the inefficiencies in booking and communication still result in noticeable revenue loss and reduced operational effectiveness.

The primary financial impacts are as follows:

- **Unbooked Cancellation Slots:**

With approximately 5–8 cancellations per week and an estimated 60% of these slots remaining unbooked, around 3–5 court sessions per week generate no revenue. Even at a small scale, this leads to a consistent loss of potential income due to the absence of a proper rebooking mechanism.

- **Double-Booking Compensation:**

The occurrence of 2–3 double-booking incidents per week results in indirect financial losses. To resolve conflicts, the owner or staff may offer alternative time slots or informal compensation, which affects potential earnings and resource utilization.

- **Membership Lapses:**

Due to the lack of an automated reminder system, some members fail to renew their memberships on time. Although the number is relatively small compared to large clubs, this still results in missed revenue opportunities and reduced member retention.

- **Administrative Overhead:**

A considerable amount of time is spent on manual booking management, updating records, and resolving conflicts. This creates an opportunity cost, as the owner and staff could otherwise focus on improving customer service or increasing engagement with players.

Impact Category	Frequency	Est. Monthly Cost	Severity
Unbooked cancelled slots	60% of cancellations	High	High
Double-booking compensation	2-3 per week	Medium	Medium
Lapsed memberships	Occasional	Medium	Medium
Administrative overhead	Daily	Medium	Medium

3.4.3 User Dissatisfaction

Questionnaire responses collected from players of the Pemsiri Indrajith Memorial Badminton Indoor Stadium indicate noticeable dissatisfaction with the current booking experience. A majority of respondents expressed frustration with the existing semi-manual system, highlighting several usability and communication issues.

The main sources of dissatisfaction identified are as follows:

- Difficulty in contacting the owner or staff during peak hours :**
 Many players reported that it is not always easy to reach the owner by phone when booking demand is high, leading to delays and missed opportunities.
- Lack of visibility into court availability:**
 Players are unable to check available time slots on their own and must rely entirely on the owner or staff for information, which is often inconvenient and time-consuming.
- No notification for available slots:**
 A large number of respondents indicated that they are not informed when cancellations occur, even if they are interested in playing at those times.
- Booking confusion and errors:**
 Some players reported uncertainty regarding booking confirmations, especially when multiple requests are handled manually, leading to occasional misunderstandings.
- Inconvenience of manual booking methods:**
 Most players expressed dissatisfaction with the need to call or physically visit the stadium to make bookings, and showed a strong preference for a digital solution.

Overall, the findings highlight that the current system does not meet user expectations in terms of convenience, transparency, and reliability. These issues reduce user satisfaction and emphasize the need for a modern, user-friendly system with real-time access and automated communication features.

3.4.4 Scalability Issues

The current system at the Pemsiri Indrajith Memorial Badminton Indoor Stadium is not capable of supporting future growth in membership or increased booking demand. The reliance on manual processes and spreadsheet-based booking management becomes increasingly difficult to manage as the number of players grows. Even at the current scale, peak-hour operations are challenging to handle, as the owner and staff must manage multiple booking requests, cancellations, and inquiries simultaneously.

As the number of players increases, the complexity of maintaining accurate booking records and avoiding conflicts will also increase. Without a centralized and automated system, this would lead to more frequent errors such as double bookings, missed entries, and unrecorded cancellations. Additionally, the current system heavily depends on the availability of the owner or staff, making it difficult to scale operations without increasing administrative workload.

The absence of digital tools such as real-time availability tracking, automated notifications, and centralized data management limits the system's ability to handle higher demand efficiently. As a result, any growth in the number of users would require additional manual effort, which is not sustainable in the long term.

Overall, the current system lacks scalability and flexibility, making it unsuitable for supporting future expansion or improving operational efficiency.

3.4.5 Technology Limitations

The reliance on a single shared spreadsheet for booking management at the Pemsiri Indrajith Memorial Badminton Indoor Stadium creates several technological limitations and risks. There is no version control or audit trail, meaning any accidental deletion or modification of booking data cannot be easily tracked or recovered unless a manual backup exists. This increases the risk of data loss and reduces overall system reliability.

The spreadsheet is typically maintained on a single device accessible to the owner or limited staff, which restricts access to booking information. As a result, bookings cannot be checked or updated remotely, and all updates must be done manually in one place. This creates delays and limits flexibility, especially when handling multiple booking requests at the same time.

Additionally, there is no integration between booking records and other related data such as membership information. This lack of integration prevents efficient data management, as information must be checked and updated separately. It also makes it difficult to generate any meaningful reports or insights about court usage, player activity, or booking patterns.

Overall, the dependence on a basic spreadsheet system introduces risks related to data accuracy, accessibility, and scalability, highlighting the need for a more secure and integrated digital solution.

3.5 Scope and Constraints

3.5.1 System Boundaries

The scope of the proposed Badminton Club Management System encompasses the following functional modules, identified as essential through the fact-finding analysis:

- **User Management Module:** Registration, authentication, and role-based access control for members, administrators, coaches, and guests.
- **Court Booking Module:** Real-time court availability display, booking creation, modification, cancellation, and transaction locking to prevent double-bookings.
- **Notification Module:** Multi-channel (SMS, WhatsApp, push) cancellation alerts, booking confirmations, reminders, and subscription-based slot availability notifications.
- **Membership Module:** Membership plan management, expiry tracking, automated renewal reminders, and benefits administration.
- **Tournament Management Module:** Tournament creation, participant registration, automated match scheduling, bracket generation, and results display.

- **Coaching Management Module:** Coaching session scheduling, participant registration, and reminder notifications.
- **Shop and Inventory Module:** Product catalogue, online purchasing, order history, and stock level management with reorder alerts.
- **Administration and Reporting Module:** User management, revenue reporting, court utilisation analytics, and system configuration.

3.5.2 Exclusions

The following items are explicitly excluded from the system scope:

- Multi-facility management (system is designed for single-club deployment).
- Advanced financial accounting beyond revenue reporting and basic transaction records.
- Integration with external sports governing body systems or ranking databases.
- Video analysis or performance tracking for players.
- Social networking or community features beyond basic user profiles.

3.5.3 Constraints

The development and deployment of the proposed system are subject to the following constraints:

Constraint Type	Description	Implication
Budgetary	Free-tier services for academic phase; LKR 12,000-15,000/month if paid	Limited to free-tier quotas
Technological	React.js, Node.js, MySQL, React Native stack; Twilio APIs	Fixed technology choices
Human Resources	5-person student team with academic supervision	Limited development capacity
Time	30-week development schedule	Fixed delivery deadline
Regulatory	PCI DSS compliance for payments; data protection	Mandatory security standards

These constraints have been carefully considered in the requirements specification presented in Section 4, ensuring that all specified requirements are achievable within the identified limitations.

4. Requirements Specification

This section presents the feasibility considerations along with the functional and non-functional requirements for the Badminton Club Management System developed for the Pemsiri Indrajith Memorial Badminton Indoor Stadium. A feasibility study was conducted prior to requirement definition to ensure that the proposed system is technically achievable, economically viable, operationally practical, and can be completed within the given time frame. The study confirmed that the system can be implemented using available technologies, within limited cost constraints, and with strong acceptance from both staff and players.

Following the feasibility evaluation, the functional and non-functional requirements were derived from the fact-finding analysis and current system evaluation presented in the preceding sections. These requirements are organised by functional module and are stated using precise and measurable language to enable clear understanding and unambiguous verification during the testing phase.

4.1 Feasibility Study

A feasibility study was conducted to evaluate the practicality of developing and implementing the proposed Badminton Club Management System for the Pemsiri Indrajith Memorial Badminton Indoor Stadium. This study examines the system from technical, economic, operational, and schedule perspectives to ensure that the proposed solution is achievable within the given constraints.

From a **technical feasibility** perspective, the system can be successfully developed using modern and widely supported technologies such as React.js, Node.js, MySQL, and React Native. These technologies are well-documented and suitable for building scalable web and mobile applications. Integration with external services such as SMS and WhatsApp

notifications can be achieved through APIs like Twilio, making the implementation of real-time communication features technically viable. The development team possesses the required foundational knowledge and skills to work with these technologies.

In terms of **economic feasibility**, the system is considered cost-effective for the academic phase, as it will utilize free-tier cloud services for hosting and deployment. This minimizes infrastructure costs while still allowing full system functionality. Although additional costs may arise in a real-world deployment scenario, the expected benefits such as improved court utilization, reduced revenue loss from unbooked cancellations, and better member retention justify the investment.

From an **operational feasibility** standpoint, the proposed system aligns well with the needs of the stadium. The current manual system creates inefficiencies and user dissatisfaction, whereas the new system introduces automation, real-time updates, and improved communication. Both staff and players are expected to adapt easily to the system due to its user-friendly design and clear advantages over the existing process. The strong interest shown by users in features such as online booking and notification alerts further supports operational acceptance.

Regarding **schedule feasibility**, the project is achievable within the planned 30-week development timeline. The structured approach of the Waterfall model allows clear phase-by-phase progress, ensuring that requirements, design, implementation, and testing are completed systematically. With proper task allocation among the development team members, the system can be delivered within the given timeframe.

Overall, the feasibility study confirms that the proposed Badminton Club Management System is practical, achievable, and beneficial. It addresses current operational issues while remaining within the technical, financial, and time constraints of the project.

4.3 Functional Requirements

Module 1: User Management Module

- 1.1.The system shall allow users to register as members or non-member guests through the web application and mobile application.
- 1.2.The system shall authenticate users through username-password credentials with secure session management.
- 1.3.The system shall enforce role-based access control distinguishing between Administrator, Member, Coach, and Guest roles.
- 1.4.The system shall allow users to update their profile information including contact details, notification preferences, and profile photograph.
- 1.5.The system shall maintain a complete booking history for each registered user, accessible through their profile dashboard.
- 1.6.The system shall support password reset functionality via email verification, with a secure token expiry of 24 hours.
- 1.7.The system shall allow administrators to view, search, filter, and manage all user accounts from the administration dashboard.

Module 2: Court Booking Module

- 2.1.The system shall display real-time court availability for a minimum 14-day booking window, with visual indicators for available, booked, and maintenance-blocked slots.
- 2.2.The system shall prevent double-booking through database transaction locking, ensuring that concurrent booking attempts for the same slot are serialised and only one succeeds.
- 2.3.The system shall allow members to book courts in 30-minute or 60-minute increments, configurable by the administrator.

- 2.4.The system shall allow booking cancellation up to a configurable time limit (default: 2 hours before slot start time).
- 2.5.The system shall automatically update court availability status within 3 seconds of any booking creation or cancellation event.
- 2.6.The system shall enforce a maximum booking limit per member per day (default: 2 hours), configurable by the administrator.
- 2.7.The system shall support waitlist functionality, allowing members to join a queue for fully booked slots and receive priority notification if a cancellation occurs.
- 2.8.The system shall integrate with the online payment gateway to require pre-payment or deposit for booking confirmation, based on administrator configuration.

Module 3: Notification Module

- 3.1.The system shall send SMS notifications to subscribed users within 10 seconds of a court booking cancellation event, containing the date, time, and court number of the freed slot.
- 3.2.The system shall send WhatsApp notifications to subscribed users within 10 seconds of a court booking cancellation event, with content identical to the SMS notification.
- 3.3.The system shall deliver in-application push notifications to mobile app users within 5 seconds of a court booking cancellation event.
- 3.4.The system shall allow users to subscribe and unsubscribe from cancellation alerts for specific time slots, days of the week, or court types through their notification preferences.
- 3.5.The system shall send booking confirmation notifications immediately upon successful booking, via the user's preferred channel (SMS, WhatsApp, email, or push).
- 3.6.The system shall send reminder notifications 2 hours before a confirmed booking start time, via the user's preferred channel.
- 3.7.The system shall maintain a notification delivery log recording the recipient, channel, timestamp, and delivery status of each notification sent.

3.8.The system shall implement rate limiting to prevent notification flooding, with a maximum of 10 cancellation alerts per user per hour.

3.9.The system shall deliver notifications to users on the waitlist for a cancelled slot before broadcasting to general subscribers.

Module 4: Membership Module

4.1.The system shall support multiple membership categories (e.g., Standard, Premium, Student, Senior) with configurable benefits and pricing.

4.2.The system shall track membership start dates, expiry dates, and renewal history for each member.

4.3.The system shall send automated renewal reminder notifications 7 days, 3 days, and 1 day before membership expiry, via the member's preferred channel.

4.4.The system shall apply membership benefits automatically during booking, including discounted rates and priority booking windows where applicable.

4.5.The system shall generate a digital membership card with a unique QR code for each active member, accessible through the mobile application.

4.6.The system shall allow administrators to create, modify, and deactivate membership plans through the administration dashboard.

4.7.The system shall provide a membership analytics report showing active members, expiring memberships, and new enrolments over a configurable time period.

Module 5: Tournament Management Module

5.1.The system shall allow administrators to create tournaments specifying name, category, format (singles/doubles), maximum participants, entry fee, and schedule.

5.2.The system shall allow members to register for tournaments and pay entry fees through the integrated payment gateway.

5.3.The system shall automatically generate tournament draws and match schedules using configurable seeding algorithms.

- 5.4.The system shall block court availability for tournament matches and display tournament schedules to all members.
- 5.5.The system shall allow administrators or designated officials to record match results and automatically advance winners through the bracket.
- 5.6.The system shall display tournament brackets, match schedules, and results through the web and mobile interfaces.
- 5.7.The system shall send tournament-related notifications including registration deadlines, match reminders, and result announcements to participants.

Module 6: Coaching Management Module

- 6.1.The system shall display coaching session schedules with coach profiles, specialisation areas, session times, and availability.
- 6.2.The system shall allow members to register for coaching sessions and make payments through the integrated payment gateway.
- 6.3.The system shall enforce maximum participant limits per coaching session and maintain a waiting list when sessions are full.
- 6.4.The system shall send coaching session reminders to registered participants 24 hours before the session.
- 6.5.The system shall allow coaches to view their schedules and registered participants through a dedicated coach portal.
- 6.6.The system shall allow administrators to create and modify coaching schedules, assign coaches, and configure session parameters.

Module 7: Shop and Inventory Module

- 7.1.The system shall display a product catalogue with product name, description, price, images, and stock availability for all items in the pro-shop inventory.
- 7.2.The system shall allow members and guests to add products to a shopping cart and complete purchases through the integrated payment gateway.

- 7.3.The system shall maintain real-time inventory levels, decrementing stock quantities upon order completion and preventing overselling.
- 7.4.The system shall send automated low-stock alerts to administrators when inventory levels fall below configurable reorder thresholds.
- 7.5.The system shall generate order history for each user and allow administrators to view and manage all orders.
- 7.6.The system shall support product categorisation, search functionality, and promotional discount codes.

Module 8: Administration and Reporting Module

- 8.1.The system shall provide a secure administrator dashboard with role-based access to all management functions.
- 8.2.The system shall generate revenue reports summarising booking income, membership fees, tournament entry fees, coaching fees, and shop sales over configurable date ranges.
- 8.3.The system shall produce court utilisation analytics showing occupancy rates by day, time slot, and court, with visual chart representation.
- 8.4.The system shall provide user management capabilities including account creation, modification, deactivation, and role assignment.
- 8.5.The system shall maintain an audit log of all administrative actions including data modifications, configuration changes, and financial transactions.
- 8.6.The system shall allow administrators to configure operational parameters including booking rules, cancellation policies, notification settings, and business hours.
- 8.7.The system shall export reports in CSV and PDF formats for external analysis and record-keeping.

4.4 Non-Functional Requirements

4.4.1 Performance Requirements

NFR-1.The system shall serve web pages with a response time not exceeding 3 seconds under normal operating conditions (up to 50 concurrent users).

NFR-2.The system shall process booking transactions with a response time not exceeding 2 seconds from submission to confirmation.

NFR-3.The notification system shall deliver cancellation alerts within 10 seconds of the triggering event for SMS and WhatsApp channels, and within 5 seconds for push notifications.

NFR-4.The system shall support a minimum of 100 concurrent users without degradation of response time below the specified thresholds.

NFR-5.Database queries for court availability shall execute within 500 milliseconds under all load conditions.

4.4.2 Security Requirements

NFR-6.The system shall encrypt all user passwords using bcrypt hashing with a minimum work factor of 10.

NFR-7.The system shall implement HTTPS encryption for all client-server communications.

NFR-8.The system shall enforce secure session management with automatic timeout after 30 minutes of inactivity.

NFR-9.The system shall comply with the Payment Card Industry Data Security Standard (PCI DSS) for all payment processing.

NFR-10.The system shall implement role-based access control preventing unauthorised access to administrative functions.

NFR-11.The system shall protect against common web application vulnerabilities including SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF).

4.4.3 Usability Requirements

NFR-12.The web application shall be responsive and fully functional on screen sizes from 320px (mobile) to 1920px (desktop).

NFR-13.The mobile application shall be available on both Android (API level 26+) and iOS (iOS 14+) platforms.

NFR-14.The system shall achieve a System Usability Scale (SUS) score of 70 or above in User Acceptance Testing.

NFR-15.The system shall provide clear error messages in plain language for all validation failures and system errors.

NFR-16.The system shall conform to WCAG 2.1 Level AA accessibility standards for the web application.

4.4.4 Availability and Reliability Requirements

NFR-17.The system shall maintain 99.5 percent uptime availability, excluding scheduled maintenance windows.

NFR-18.The system shall implement automated database backups at minimum 4-hour intervals, with a recovery point objective (RPO) of 4 hours.

NFR-19.The system shall recover from service failures automatically within 5 minutes without data loss.

NFR-20.The system shall maintain accurate and consistent booking records with zero tolerance for data loss or corruption.

4.4.5 Scalability and Maintainability Requirements

NFR-21.The system architecture shall support horizontal scaling to accommodate membership growth to 1,000 active users without major redesign.

NFR-22.The system shall allow addition of new courts and modification of operating hours through configuration changes without code modification.

NFR-23.The system shall be developed using modular architecture principles, enabling independent deployment and modification of functional modules.

NFR-24.The system shall maintain comprehensive API documentation to support future integration with external services.

4.5 System Constraints

The following constraints apply to the system design and implementation:

SC-1.The system shall be developed using the Waterfall Software Development Life Cycle model, with sequential completion of requirements, design, implementation, testing, and deployment phases.

SC-2.The system shall utilise React.js for the web frontend, Node.js with Express.js for the backend, MySQL for the database, and React Native for the mobile application.

SC-3.The system shall integrate with Twilio API for SMS notifications and WhatsApp Business API for WhatsApp notifications.

SC-4.The system shall be hosted on free-tier cloud services (Vercel for frontend, Render for backend, Railway for database) for the academic phase, with architecture supporting migration to production hosting.

SC-5.The system shall be delivered within a 30-week development schedule as specified in the project plan.

SC-6.The system development shall be conducted by a five-person student team with academic supervision.

These requirements collectively define the complete functional and quality specification for the Badminton Club Management System. All requirements shall be verified through the structured testing methodology outlined in the project methodology document, encompassing unit testing, integration testing, system testing, and user acceptance testing phases.

5. References

- “What is an Analysis Report & How to Create it: Templates & Samples”
<https://www.finereport.com/en/reporting-tools/analysis-report.html>
- “Writing a Good Data Analysis Report :7 Steps”
<https://www.modernanalyst.com/Resources/Articles/tabid/115/ID/6070/Writing-a-Good-Data-Analysis-Report-7-Steps.aspx>
- “Best Practices for Requirements Gathering in Project Management”
<https://www.runn.io/blog/requirements-gathering>
- “Functional vs. Non-Functional Requirements”
https://qracorp.com/guides_checklists/functional-vs-non-functional-requirements/
- Modern Systems Analysis and Design Valacich, J. S., George, J. F., & Hoffer, J. A. (2017). Modern Systems Analysis and Design (8th ed.). Pearson.
<https://www.pearson.com/store/p/modern-systems-analysis-and-design/P200000011446/9780138180195>
- • IEEE – IEEE 29148 (SRS Standard) IEEE (2018). ISO/IEC/IEEE 29148: Systems and software engineering Life cycle processes Requirements engineering.
https://www.iso.org/standard/72089.html?utm_source=chatgpt.com

